

Youhao Hu

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Personal Information

Date of Birth: 16 August 1994

Address: 1st Duxue Road, Nansha District, Guangzhou, Guangdong, China, 511400

Education

Ph.D. The Hong Kong University of Science and Technology (Guangzhou), Power Electronics Guangzhou, China
Sept 2022 – present

- Supervisors: Dr. Wei Han and Prof. C. C. Chan.

M.S. Hefei University of Technology, Control Engineering Hefei, China
Sept 2017 – June 2020

B.S. Shandong University of Technology, Automation Zibo, China
Sept 2013 – June 2017

Research and Engineering Experience

Wireless-Powered Full-Azimuth Podded Thruster, New Propulsion System for Unmanned Vessels Guangzhou, China

- Developed a wireless-powered propulsion prototype for unmanned vessels, targeting sealing reliability and reduced maintenance compared with conventional podded propulsion solutions.
- Modeled and designed wireless power transfer system parameters.
- Developed control and integration for a high-frequency wireless power converter.
- Implemented motor drive control, including field-oriented control and real-time control.
- Contributed sealed mechanical structure design, 3D modeling, prototype integration, debugging, and validation.
- Built a system including a 1200 W / 100 V GaN motor driver, a 5 kW integrated WPT inverter, 5-10 cm transfer distance, and 92% peak efficiency.

Active-Rectifier Wireless Motor System, Research Highlight

- Developed system-level analysis and dual-side phase shift control for active-rectifier wireless motor systems.
- Related selected paper: An Active-Rectifier Wireless Motor System with Dual-Side Phase Shift Control.

Fixed-Time Voltage Regulation for Wireless Power Transfer, Research Highlight

- Combined neural-network-assisted control with adaptive practical fixed-time voltage regulation for wireless power transfer systems.
- Related selected paper: ELM-Based Adaptive Practical Fixed-Time Voltage Regulation for Wireless Power Transfer System.

Electronic Throttle Control, Research Highlight

- Developed adaptive finite-time and fixed-time sliding mode control algorithms for electronic throttle control systems.
- Addressed nonlinear dynamics, robustness, convergence speed, and learning-assisted compensation.

Skills

Power Electronics System Design: Power electronics modeling, wireless power transfer, wireless motor systems, bidirectional energy conversion

Complex Dynamics Modeling: System modeling, nonlinear dynamics, dynamic wireless power transfer

Sliding Mode Control Theory: Adaptive control, barrier-function control, finite-time control, fixed-time control

Embedded Systems Programming: Matlab, Python, C, Windows development, embedded systems

Neural Network Methods: Extreme learning machine, neural-network-assisted control, adaptive approximation

Honors and Awards

2026: Gold Medal, International Exhibition of Inventions of Geneva, Palexpo SA, Switzerland

2022: Fully Funded PhD Scholarship, The Hong Kong University of Science and Technology (Guangzhou), China

2020: Outstanding Graduate Award, Anhui Province, China

2020: Outstanding Graduate Award, Hefei University of Technology, China

2019: National Scholarship for Graduate Student, Ministry of Education, China

2019: Academic First Grade Scholarship, Hefei University of Technology, China

2018: Academic Second Grade Scholarship, Hefei University of Technology, China

2017: Academic First Grade Scholarship, Hefei University of Technology, China

2017: Outstanding Undergraduate Award, Shandong University of Technology, China

2016: Meritorious Winner, Interdisciplinary Contest in Modeling, COMAP, United States

2016: Qingdao DORIGHT Scholarship, Shandong University of Technology, China